Counting methods

1. The sum rule (the rule of "or")

Obs. If a counting problem is partitioned into mutually exclusive (disjoint) cases which cover all possible outcomes, the total number of outcomes is the sum of the outcomes of each case.

Def. (The sum rule) For any finite sets $A_1, A_2, \ldots, A_m$, pairwise disjoint ($A_i \cap A_j = \emptyset \ \forall i \neq j$):

$$|A_1 \cup A_2 \cup A_3 \cup \ldots \cup A_m| = |A_1| + |A_2| + \ldots + |A_m|$$

Obs. The cases must not overlap. When cases overlap, we apply other methods, such as the inclusion-exclusion principle.

ex. Counting even three-digit numbers with distinct digits:

Case 1: $C = 0 \Rightarrow \overline{9}\ \overline{8}\ \overline{1} = 9 \cdot 8 = 72$

Case 2: $C \neq 0 \Rightarrow \overline{8}\ \overline{8}\ \overline{4} = 8 \cdot 8 \cdot 4 = 256$

$\Rightarrow S = 328$

2. The difference rule (The complement rule)

Def. (The complement rule) For any finite set U and any subset $A \subseteq U$:

$$|U \setminus A| = |U| - |A|$$

Obs. The number of objects satisfying a given property equals the total number of objects minus those which do not satisfy the property.

Obs. Effective for questions involving "at least one".

ex. How many three-digit numbers contain the digit 7?

900 - total numbers
$\overline{8}\,\overline{9}\,\overline{9}$ = 648 without 7 $\Big| \Rightarrow$ 252 nr. with 7

3. The general strategy

Organize complex counting tasks using a Divide and Conquer approach:

1. Divide into smaller, similar subproblems.
2. Solve each problem using the appropriate rule (bijection, product, sum, complement etc.)
3. Merge the sub-results.

4. Classification of choices (k objects from n distinct objects)

	without repetition	with repetition
Order matters	k-permutation	k-perm with rep.
Order doesn't matter	combinations	combinations with rep.

5. k-permutations without repetition (linear)

Def. An ordered sequence of k distinct objects chosen from n distinct objects ($n \geq k$):

$$n^{\underline{k}} = n(n-1)\dots(n-k+1) = \frac{n!}{(n-k)!}$$

(if $k > n \Rightarrow n^{\underline{k}} = 0$)

Obs. Permutations ($k = n$) are the number of ways to arrange all n distinct objects in a row:

$$n^{\underline{n}} = n!$$

6. k-permutations with repetition:

Def. An ordered sequence of k objects chosen from n distinct types, where elements can repeat:

$$n^k$$

7. Counting functions:

Def. Given finite sets X ($|X| = k$) and Y ($|Y| = n$):

1) Total functions $f: X \to Y = n^k$

2) Injective functions $f: X \to Y = n^{\underline{k}}$

3) Bijective functions $f: X \to Y = n!$

8. Circular k-permutations (without repetition)

Def. Arranging k distinct objects from n available objects around a circle, where rotations produce equivalent arrangements:

1) Circular k-permutations ($n \geq k \geq 1$):

$$\frac{n^{\underline{k}}}{k}$$

2) Circular permutations:

$$\frac{n!}{n} = (n-1)!$$

3) Round table with empty seats (n people with k seats):

$$(K-1)^{\underline{n-1}} = \frac{k^{\underline{n}}}{k}$$